

Hotel Booking Analytics & Revenue Optimization

Data-Driven Insights using Python, SQL, Power BI And Orange

Prepared by:

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Dataset

Project Codes

Hotel Booking Data Analysis Report

1. Abstract

This report presents an exploratory data analysis (EDA) of the Hotel Booking Demand dataset. The dataset contains booking information for City Hotels and Resort Hotels between July 2015 and August 2017. The purpose of the analysis is to identify booking patterns, customer behavior, cancellation trends, and seasonal demand.

Insights from this analysis can help hotel managers improve revenue management, pricing strategies, and customer retention. The analysis includes data cleaning, preprocessing, visualization, and interpretation of patterns within the data. Key findings indicate that city hotels receive a higher number of bookings compared to resort hotels, online travel agencies contribute significantly to reservations, and long booking lead times increase the likelihood of cancellations. These insights provide practical recommendations for improving operational efficiency and reducing revenue loss.

2. Introduction

The hospitality industry is highly competitive and data-driven decision-making is becoming increasingly important for hotel management. Hotels collect large volumes of booking data, which can be analyzed to understand customer preferences, seasonal demand patterns, and operational challenges. One of the major challenges hotels face is booking cancellations. Cancellations can significantly affect revenue, management, staffing, and room availability planning. By analyzing historical booking data, hotels can identify patterns that lead to cancellations and implement strategies to mitigate these risks.

This project analyzes the Hotel Booking dataset available on Kaggle. The dataset contains booking information for two types of hotels: City Hotel and Resort Hotel. The analysis focuses on understanding booking patterns, seasonal trends, customer segments, and cancellation behavior.

The insights generated from this analysis can help hotel managers improve pricing strategies, reduce cancellations, and enhance customer satisfaction.

3. Dataset Description

The dataset used in this project contains booking information for two hotel types: City Hotels and Resort Hotels. The data covers bookings made between July 2015 and August 2017.

Dataset characteristics:

- Number of records: approximately 119,390
- Number of features: 32
- Time range: July 2015 to August 2017

Important variables include:

- hotel: type of hotel (City or Resort)
- is_canceled: booking cancellation status
- lead_time: number of days between booking and arrival
- arrival_date_year and arrival_date_month
- number of adults, children, and babies
- meal type selected
- market segment and distribution channel
- average daily rate (ADR)
- customer type

These features allow the analysis of booking behavior, guest demographics, and revenue patterns.

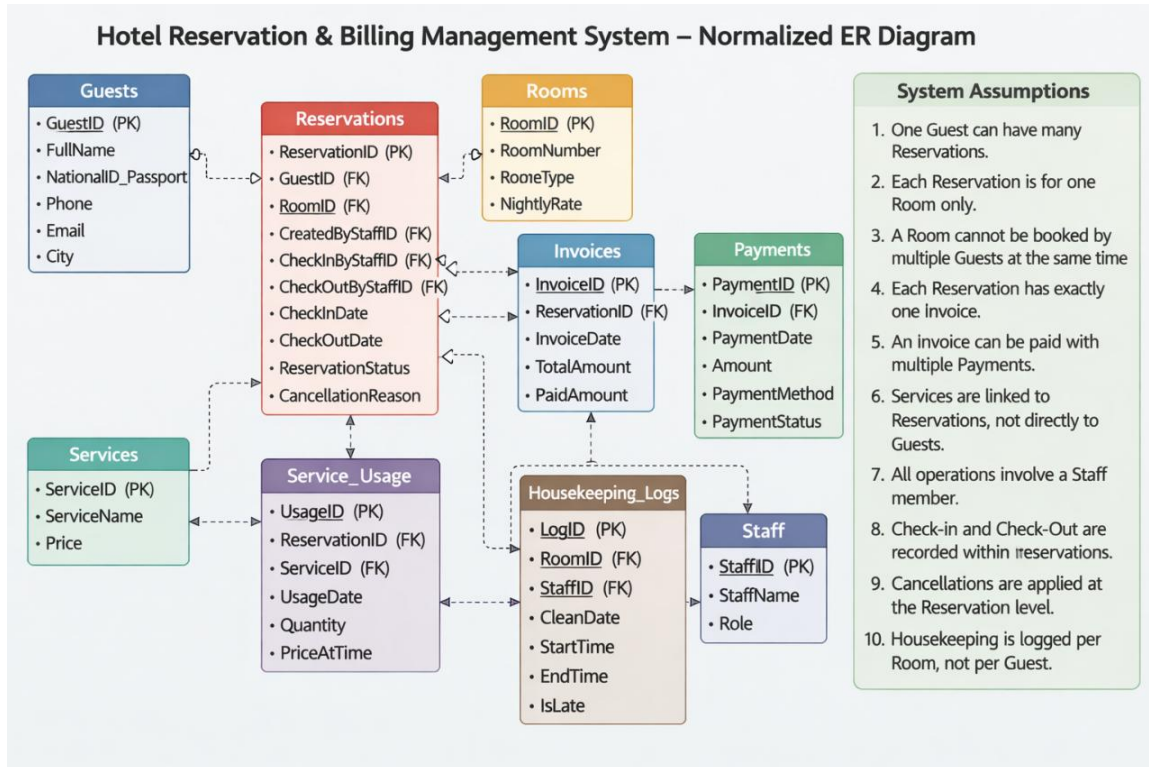
4. Project Objectives

The main goals of this analysis are:

1. Analyze booking trends for City and Resort hotels.
2. Identify factors affecting booking cancellations.
3. Explore seasonal patterns in hotel reservations.
4. Understand customer demographics and booking behavior.
5. Provide actionable recommendations for hotel management.

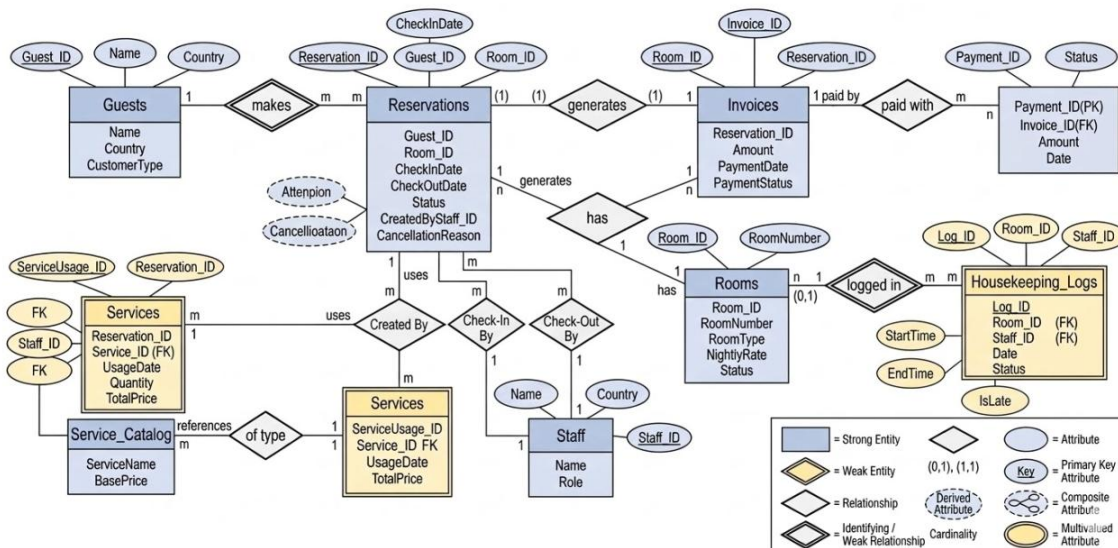
5. Database Design Section

The database schema illustrates the structure of the dataset and the relationships between tables used in the analysis



ERD

Hotel Reservation & Billing System – ER Diagram (Chen Notation)



ERD Explanation

1. Entities

Entities represent the main objects or concepts stored in the database. Each entity corresponds to a table that contains related information. In the hotel booking system, the main entities include:

Booking: This entity stores the main reservation information such as booking status, arrival date, length of stay, and average daily rate (ADR).

Customer: This entity contains guest-related information such as the number of adults, children, babies, and customer type (transient, contract, group, etc.).

Hotel: The hotel entity identifies the hotel category where the reservation was made, such as **City Hotel** or **Resort Hotel**.

Market Segment: This entity describes the type of market from which the booking originated, such as Online Travel Agency (OTA), Direct booking, or Corporate clients.

Distribution Channel: This entity indicates how the booking was distributed or reserved, for example through travel agents, direct booking, or online platforms.

These entities help organize the data into logical groups that make analysis easier.

2. Relationships

Relationships describe how the entities are connected to each other within the database

In this system:

- A **hotel** can have **many bookings**
- A **customer** can make **multiple bookings**
- Each **booking belongs to one hotel**
- Each **booking is associated with one market segment**
- Each **booking uses one distribution channel**

These relationships allow analysts to connect booking data with customer information and hotel characteristics, enabling deeper insights into booking behavior and market trends.

3. Primary Keys

A primary key is a unique identifier used to distinguish each record in a table. It ensures that every row in the table can be uniquely identified.

Examples of primary keys in the system include:

Booking_ID – uniquely identifies each reservation

Customer_ID – uniquely identifies each customer

Hotel_ID – uniquely identifies each hotel

Market_Segment_ID – uniquely identifies each market segment

Channel_ID – uniquely identifies each distribution channel

Primary keys prevent duplicate records and ensure the integrity of the database.

4. Foreign Keys

Foreign keys are fields in a table that reference the primary key of another table. They are used to establish relationships between tables.

Examples include:

Hotel_ID in the Booking table → references the Hotel table

Customer_ID in the Booking table → references the Customer table

Market_Segment_ID in the Booking table → references the Market Segment table

Channel_ID in the Booking table → references the Distribution Channel table

Foreign keys help link related data across tables and allow complex queries to retrieve combined information from multiple entities.

6. Data Preprocessing

Before performing analysis, several preprocessing steps were applied to ensure data quality.

Handling missing values:

Some columns contained missing values such as:

Children, country, agent, company.

Missing values were handled by:

- removing columns with excessive missing data.
- Some columns were merged and those containing missing values were cleared.

7. Exploratory Data Analysis

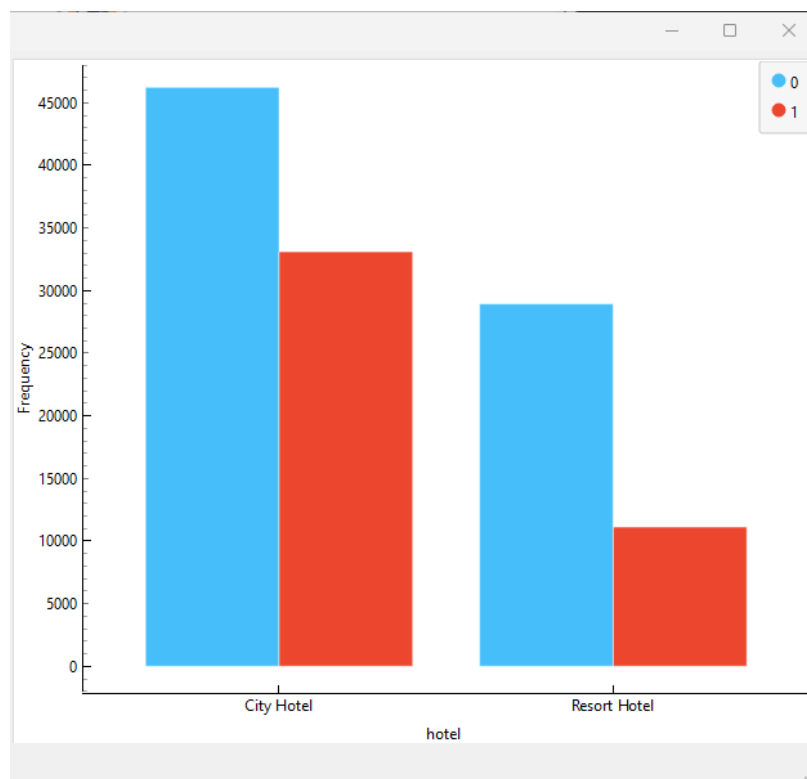
Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset and identify meaningful patterns.

Distribution of hotel types:

The dataset shows that City Hotels receive more bookings than Resort Hotels. This may be due to higher business travel demand in urban areas.

Booking cancellations:

A large percentage of bookings are canceled. Lead time plays a major role in cancellations, as bookings made far in advance have a higher probability of being canceled.

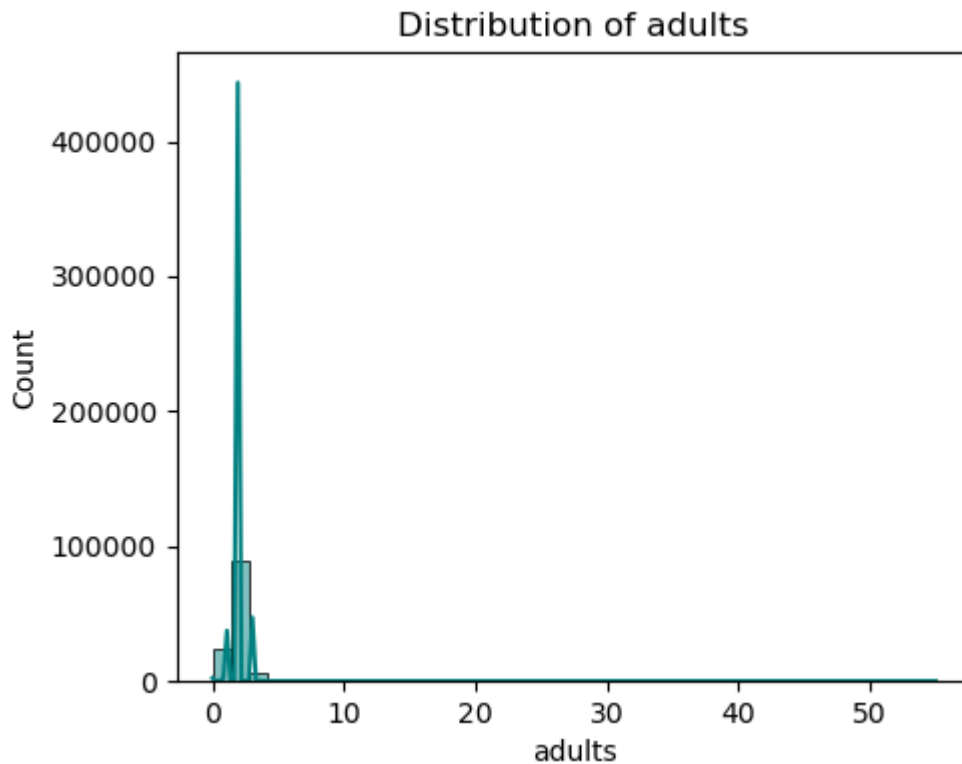


Seasonal trends:

Booking demand varies across months. Summer months such as July and August show higher booking activity, especially for resort hotels.

Customer demographics:

Most bookings involve two adults, Families with children represent a smaller portion of the Dataset The majority of guests are transient customers rather than long-term guests.



8. Booking Behavior Analysis

Booking behavior analysis helps understand how customers interact with hotel reservation systems.

Lead time analysis:

Customers often book rooms several weeks or months in advance Longer lead times tend to correlate with higher cancellation rates.

Monthly Booking Trends:

Seasonal demand patterns were identified:

Peak months: July, August, and summer seasons

Lower demand: Winter months

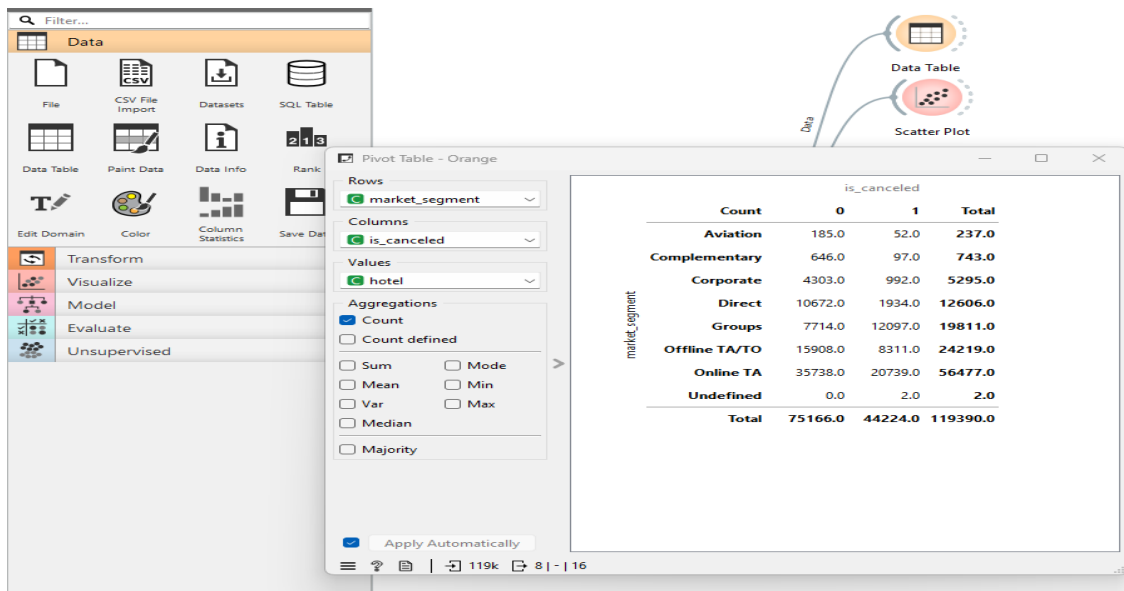
Resort hotels experience strong seasonality due to tourism patterns.

Length of stay:

Most guests stay between one and three nights Longer stays are less common but are more frequent in resort hotels.

Distribution channels:

Online travel agencies (OTAs) represent the largest distribution channel Direct bookings also represent a significant portion but remain smaller than OTA bookings.



Customer Demographics

Key insights include:

Most bookings involve **2 adults**.

Families with children represent a smaller proportion.

The majority of customers are **transient travelers** rather than long-term guests.

Average Daily Rate (ADR)

ADR analysis reveals:

Resort hotels often have higher ADR during peak seasons.

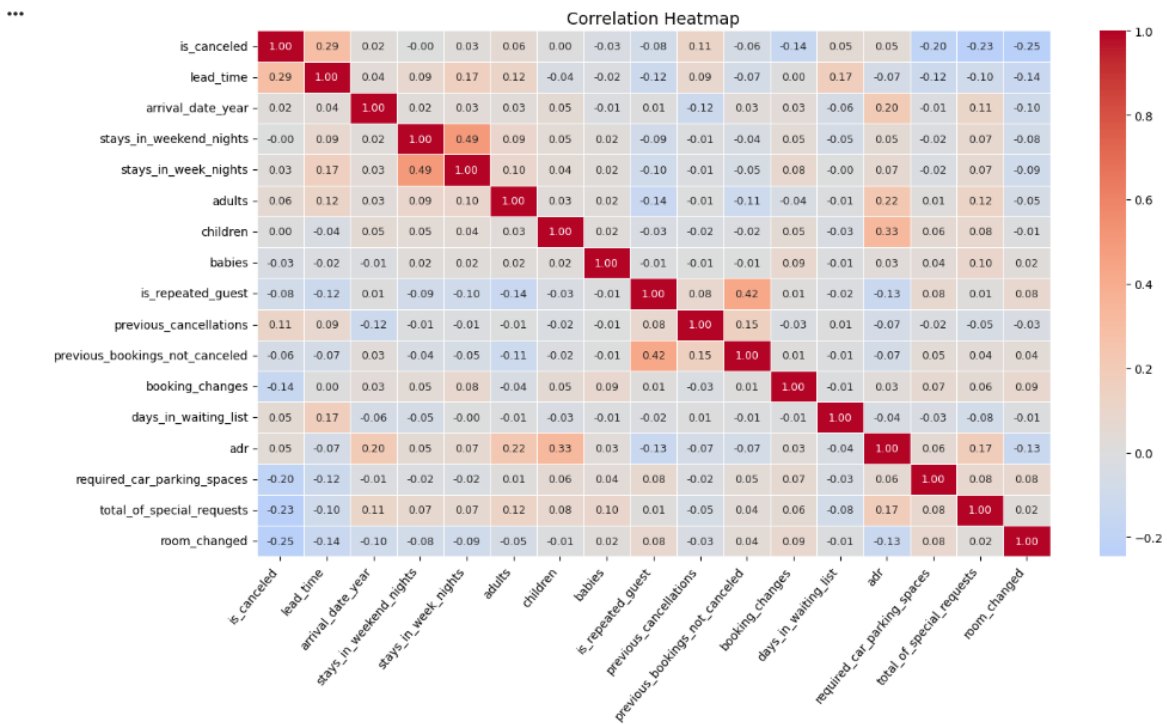
Pricing fluctuates depending on demand and seasonality.

The Cancellation Analysis:

Lead Time: (0.29) is our biggest risk—the further out they book, the more likely they cancel. Conversely, guests who make Special Requests (-0.23) or ask for Parking (-0.20) are our "stickiest" and most reliable customers.

The Family Revenue Engine: There is a clear "Family Cluster" where Children (0.33) and Adults (0.22) drive our highest ADR. Families aren't just guests; they are our primary high-yield revenue drivers.

The Loyalty Loop: Repeat Guests show a strong (0.42) correlation with Previous Bookings Not Canceled. These are our lowest-risk assets; we should offer them the most flexible terms to keep them bypassing the OTAs.



9. Pricing Analysis

Average Daily Rate (ADR) represents the average price paid for rooms.

Key findings:

- Resort hotels generally have higher ADR during peak tourist seasons.
- City hotel prices are more stable throughout the year.
- ADR increases during high demand periods such as summer months and holidays.

Understanding pricing trends allows hotels to implement dynamic pricing strategies that maximize revenue while remaining competitive.

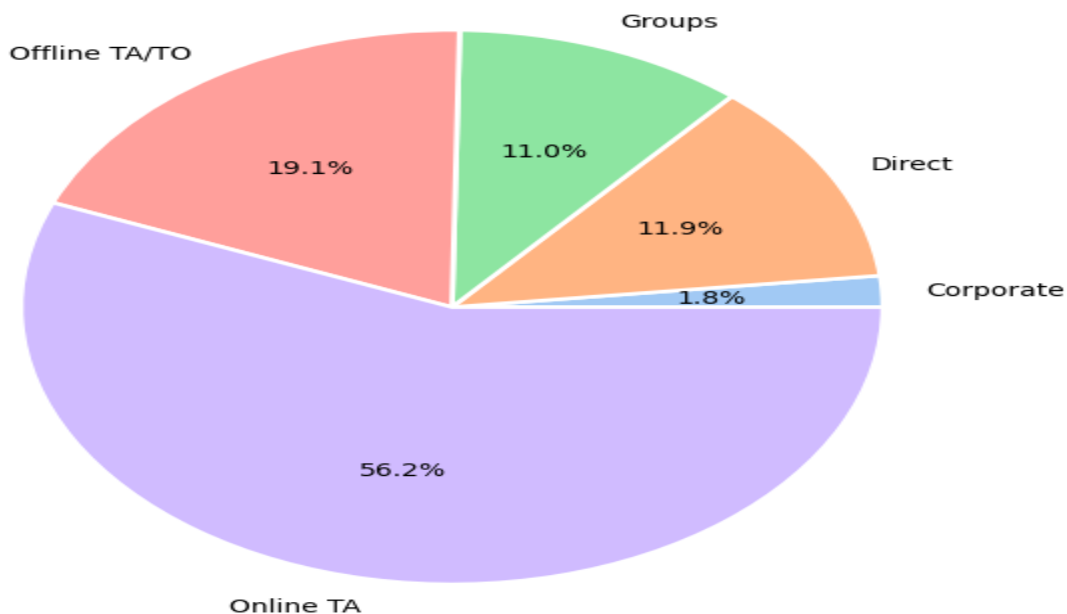
10. Revenue Dominance

Revenue Dominance: Online TA is the primary revenue driver at 56%, followed by Offline TA/TO at 19%. Small segments (Aviation, Corporate) were excluded from this view as their financial impact is negligible (near 0%).

Action:

Since Online TA controls over half our revenue, negotiate better commission rates or shift 5-10% of this volume to Direct channels to save on fees.

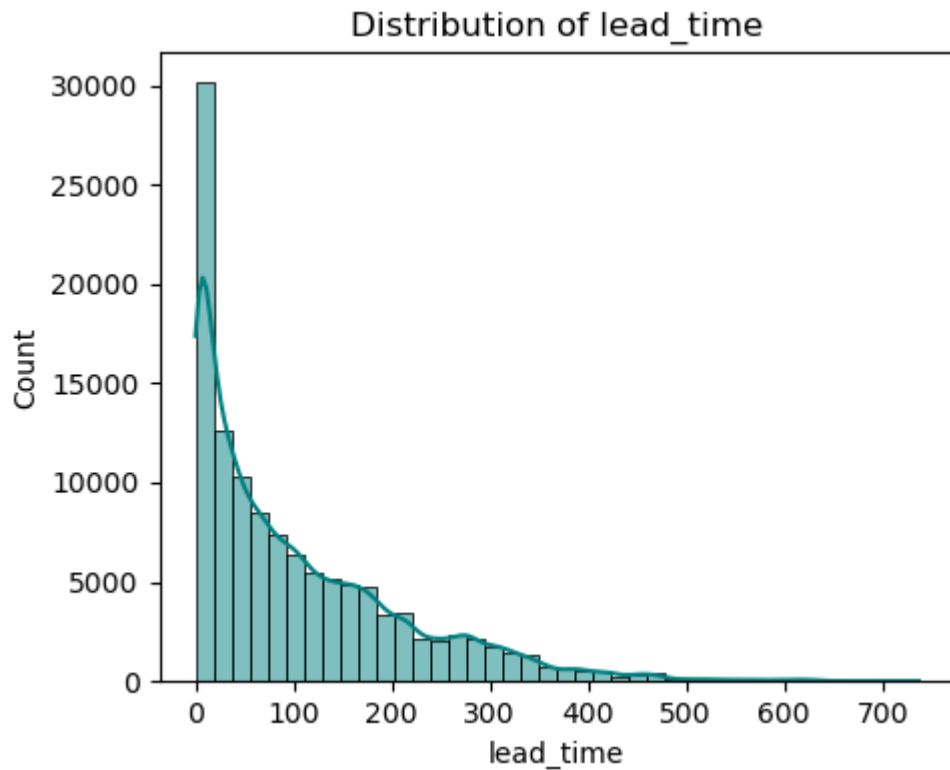
Major Revenue Contributors (>1% Share)



11. Key Insights

Several important insights were discovered during the analysis:

1. City hotels receive more bookings compared to resort hotels.
2. Long lead times are associated with higher cancellation rates.



3. Summer months generate the highest booking demand.
4. Online travel agencies dominate booking channels.
5. Most customers stay for short durations of one to three nights.
6. The majority of guests are transient travelers.

These findings provide valuable information for hotel managers seeking to improve operational efficiency.

12. Model Evaluation and Performance Comparison

Figure X: Evaluation Results of Machine Learning Models

To evaluate the predictive performance of the models, a 5-fold stratified cross-validation method was applied. This technique divides the dataset into five subsets and trains the model multiple times to ensure reliable and unbiased results.

Three classification algorithms were tested:

(Logistic Regression , Random Forest ,Decision Tree)

The evaluation was based on several performance metrics including AUC (Area Under the ROC Curve), Classification Accuracy (CA), F1 Score, Precision, Recall, and Matthews Correlation Coefficient (MCC).

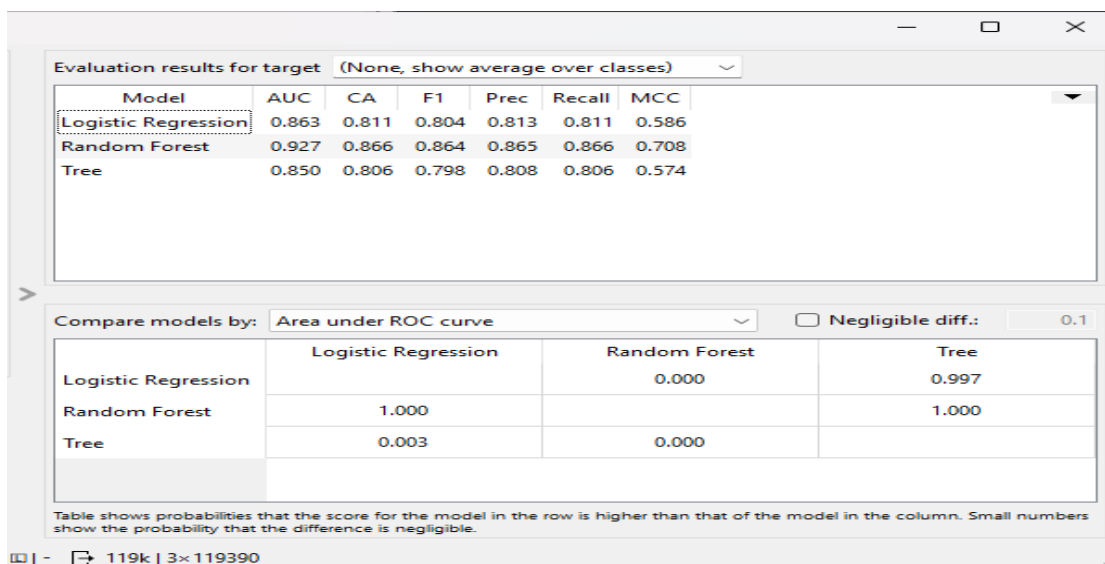
The results show that **Random Forest** achieved the best overall performance, with an AUC of 0.927 and accuracy of 0.866, indicating a strong ability to correctly classify booking outcomes. It also obtained the highest values for F1 score (0.864), precision (0.865), recall (0.866), and MCC (0.708).

Logistic Regression also demonstrated good performance with an AUC of 0.863 and accuracy of 0.811, showing stable and balanced classification results.

Meanwhile, **the Decision Tree** model achieved slightly lower results with an AUC of 0.850 and accuracy of 0.806, but it still provided reasonable predictive capability.

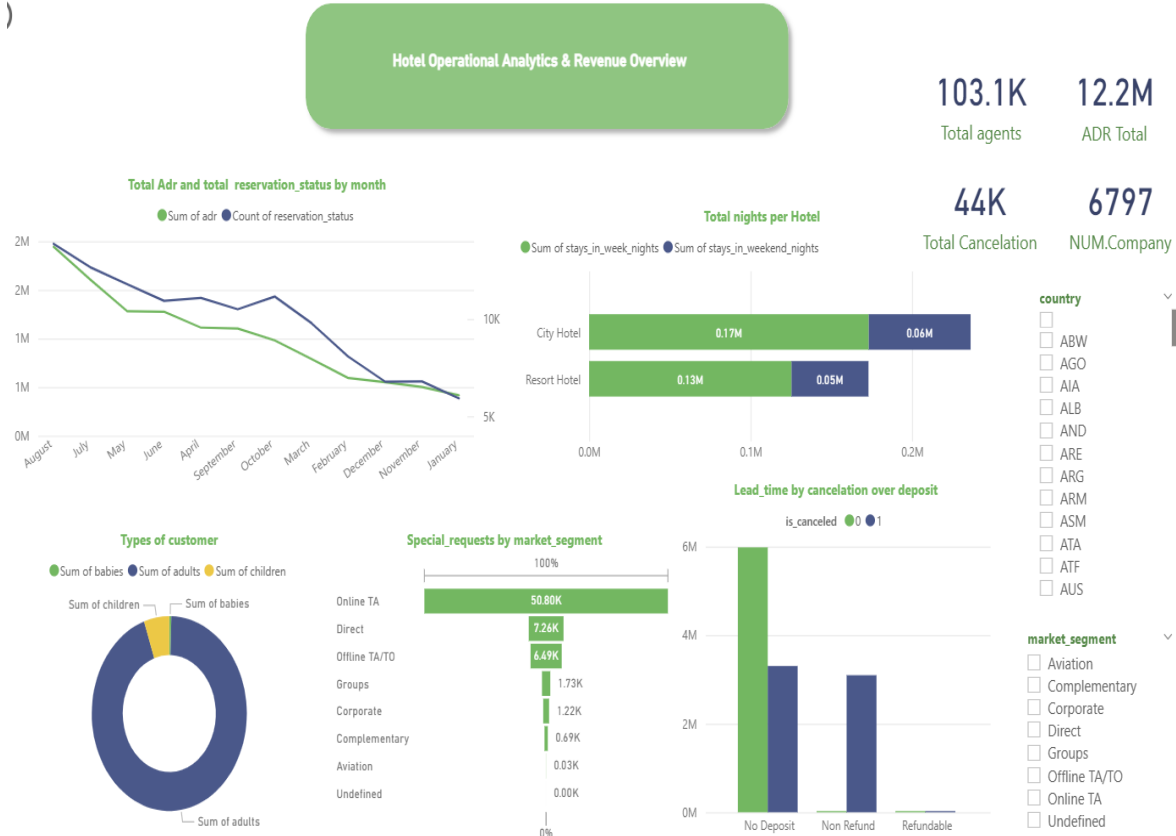
Overall, the comparison indicates that Random Forest is the most effective model for this dataset, as it consistently achieved the highest scores across most evaluation metrics.

Figure X: Performance comparison of Logistic Regression, Random Forest, and Decision Tree models using 5-fold cross-validation



13. Power BI Dashboard Analysis

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To better analyze hotel booking performance and operational trends, an interactive dashboard was developed using Microsoft Power BI. The dashboard provides a comprehensive overview of booking patterns, revenue indicators, customer characteristics, and cancellation behavior.

The dashboard displays several key performance indicators that summarize the hotel booking dataset. These include the total number of agents, total cancellations, total companies involved in bookings, and the overall average daily rate (ADR). These metrics provide a quick overview of the operational performance of the hotel booking system.

The monthly trend visualization compares the **total ADR and the number of reservations across months**. This chart helps identify seasonal booking patterns and highlights periods of higher or lower demand. Another important visualization shows the **total number of nights spent in City Hotels and Resort Hotels**, distinguishing between weekday and weekend stays. The chart indicates that City Hotels generally receive a higher number of total nights compared to Resort Hotels.

The dashboard also analyzes **customer composition**, displaying the distribution of adults, children, and babies in hotel reservations. The visualization shows that the majority of bookings are made by adults, while bookings including children or babies represent a smaller proportion. In addition, the dashboard includes an analysis of **special requests by market segment**. The results indicate that the **Online Travel Agency (Online TA)** segment generates the largest number of requests compared to other segments such as Direct, Offline Travel Agents, or Corporate bookings. Another important component of the dashboard is the **lead time and cancellation analysis**. This chart illustrates how different deposit types (No Deposit, Non Refund, and Refundable) relate to booking cancellations, helping identify patterns that may influence cancellation behavior.

Finally, interactive filters such as **country and market segment** allow users to explore the data dynamically and focus on specific regions or booking channels.

Overall, the Power BI dashboard transforms raw booking data into clear and interactive visual insights, enabling better understanding of hotel operations and supporting data-driven decision making.

14. Business Recommendations

Based on the insights obtained from the analysis, several recommendations can be made:

Improve cancellation policies:

Hotels can reduce cancellation losses by applying stricter cancellation rules for bookings made far in advance.

Encourage direct bookings:

Hotels should invest in marketing campaigns and loyalty programs to encourage customers to book directly.

Implement dynamic pricing:

Adjust room prices based on seasonal demand and booking trends.

Focus on customer retention:

Offering loyalty programs and personalized offers can increase repeat bookings.

Use predictive analytics:

Machine learning models can be developed to predict cancellations and demand trends.

15. Tools and Technologies

The following tools were used during this project:

SQL – Database Creation.

ORANGE- For modeling.

Python – data analysis programming language.

Google Colab – development environment.

Pandas – data manipulation and analysis.

NumPy – numerical computing.

Matplotlib and Seaborn – data visualization.

Power BI- Dashboard.

These tools enabled efficient analysis and visualization of the dataset.

16. Conclusion

This project demonstrates how data analysis can uncover valuable insights within hotel booking data. By analyzing booking trends, cancellation patterns, customer behavior, and pricing strategies, hotels can make more informed business decisions.

The findings highlight the importance of managing cancellations, understanding seasonal demand, and optimizing distribution channels. Future work could include developing machine learning models to predict booking cancellations and demand forecasting systems to support revenue management. Data-driven decision-making will continue to play an essential role in improving hotel operations and maximizing profitability in the hospitality industry.